

A mechanistic approach for predicting mass transfer in bioreactors

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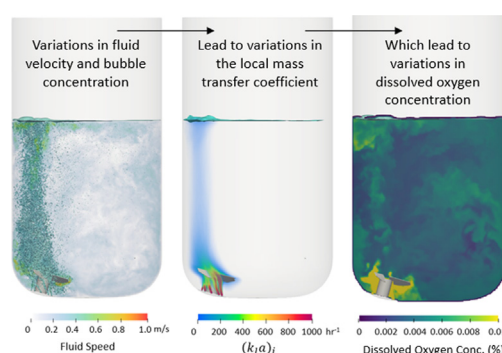
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HIGHLIGHTS

- Oxygen transfer in bioreactors is governed by complex fluid mechanics.
- These complexities make predictive mathematical modeling slow and difficult.
- Here we develop a general, physics-based approach to modeling oxygen transfer.
- We purposely design the model to execute on fast-running GPU-based computers.
- The models run fast and correctly predict oxygen transfer at multiple vessel scales.

GRAPHICAL ABSTRACT



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ABSTRACT

The biomanufacturing processes that produce biologic drugs has become an extremely important area of study within the pharmaceutical industry. Within such processes, the drug substance is typically produced by living organisms within stirred tank bioreactors that require a continuous supply of sparged oxygen. The overall oxygen transfer rate to the fluid is a nonlinear convolution of the gas bubble size distribution, fluid properties, local fluid energy dissipation rates, and local dissolved oxygen concentrations. The complexity of this process presents challenges to process scale-up and intensification. In this work, we propose, implement, and validate a mechanistic transport model for predicting oxygen transfer rates within stirred tank bioreactors. To begin, we describe the relevant conservation laws and key principles from turbulence theory that govern mass transfer. Next, we present a physics-based modeling approach for solving these equations in tandem and in real-time. We then systematically validate the model against experimental data at operating scales ranging from 5 L to 2000 L. By running the algorithm on graphics processing units (GPUs), the approach is shown to solve at timescales practical for industrial application.

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1. Introduction

During many upstream biopharmaceutical manufacturing processes, biologic drug substances are produced by living cells growing within stirred tank bioreactors (Mandenius, 2016). These bioreactors are designed to realize specific chemical/fluid environments

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Nomenclature

Latin

A	Bubble surface area
a	Constant
b	Constant
C	Constant
c	Species concentration
C_o	Courant number
C_s	Smagorinsky coefficient
D	Diffusion coefficient
d	Bubble diameter
d_h	Bubble pair harmonic mean diameter
dm	Mass of differential fluid element
$\vec{F}_a$	Added mass force
$\vec{F}_D$	Drag force
$\vec{F}_g$	Gravity force
$\vec{F}_p$	Pressure gradient force
f	Probability density function
$\vec{g}$	Acceleration due to gravity
H	Overall mass transfer coefficient or tank height
h	Depth of bubble below the free surface
k_l	Convective mass transfer coefficient
m	Bubble mass
N	Impeller speed
N_p	Non-dimensional power number
$\dot{n}_i$	Mass transfer rate from bubble
p	Pressure
P_d	Total power dissipation
P_s	Total shaft power
R_u	Universal gas constant
Re_a	Bubble pair approach Reynold number
S	Strain rate tensor magnitude or solubility

Sc	Schmidt number
T	Temperature or tank diameter
t	Time
V	Volume
$\vec{V}$	Velocity vector
V_t	Impeller tip speed
$\vec{v}$	Phase space velocity
V_m	Molar Volume
w_p	Projected blade width

Greek

Δx	Lattice spacing,
Δt	Simulation time step.
ε	Local energy dissipation rate
θ_{95}	95% Blendtime
$\nu_{r,j}$	Local eddy viscosity
ν	Kinematic viscosity.
ν_T	Total viscosity (eddy plus molecular)
ρ	Fluid density
σ	Surface tension
τ	Impeller torque or integration time window
Ω	Collision operator

Subscripts

atm	Atmospheric pressure
f	Liquid phase
g	Gas phase
i	Bubble index
j	Lattice site
R	Reference value

that maximize process yield (Doran, 1995). One key aspect of this environment is the dissolved oxygen concentration, a fluid property that impacts cell growth, metabolism, and protein quality (Van't Riet and Tramper, 1991). Maintaining a sufficient total dissolved oxygen concentration, as well as ensuring that the concentration is uniform across the tank, are key components of optimal bioreactor operation (Lewis et al., 2016).

Most biologic processes are developed and optimized in bench-top bioreactors with operating volumes smaller than 5 L. Most production-scale bioreactors, however, have volumes measured in thousands to tens-of-thousands of liters. Challenges arise when trying to scale-up the fluid/chemical environments from the lab-scale to these much-larger production-sized vessels (Paul et al., 2004). Dissolved gas concentrations are one of these key environmental parameters, and are informed by a complex convolution of the blend time, energy dissipation rate, gas injection rate, and bubble size (Lewis et al., 2016; Paul et al., 2004; Perry and Green, 2008). It is not possible, in general, to uniformly match these physics across all scales (Yang, 2011). Moreover, in vessels where the species reaction rates are fast relative to the gas transfer rates, especially in cultures with high cell densities, the dissolved gas concentrations are likely to vary across the fluid (Elqotbi et al., 2013).

Additional complexities are introduced by the proliferation of single use (SU) bioreactors as the industry moves to more intensified and flexible processes (Shukla and Gottschalk, 2013). However, since most of the literature data available for mixing systems is based on top-entry impeller configurations, the applica-

bility of this data to bottom-mounted impellers is questionable (Grenville et al., 2018). In general, bottom-mounted impellers achieve slow mixing performance compared to top-entry systems (Dreher et al., 2014). This behavior can exacerbate inhomogeneities in the dissolved oxygen transfer rates and concentrations between operating scales (Garcia-Ochoa and Gomez, 2009).

Numerical modeling is a common approach for understanding transport processes in single-phase biopharmaceutical manufacturing devices (Sharma et al., 2011). To properly capture the complexity of mammalian cell culture processes, a two-phase fluid model must be developed that is capable of handling both agitation and gassing (Dhanasekharan et al., 2005). In addition to mechanical coupling between the two phases, the model must also support species transport across the bubble/liquid interface (Bhole et al., 2008). Bubble interactions, along with changes in bubble size due to break-up and coalescence, should also be considered (Marques et al., 2010). These added requirements make two-phase bioreactor models significantly more complex than single-phase models (Delafosse et al., 2018).

The most common approach for building multi-phase bioreactor models is to combine a time-average solution to the fluid flow with a continuous (Eulerian) description of the bubbles (Haringa et al., 2017). Within the time-averaged approach, the fluid motion is solved according to the Reynolds-Averaged Navier Stokes (RANS) equations using a k - ε turbulence model (Amer et al., 2019). The bubbles are described via a continuous number density across the fluid and interactions are handled via user-calibrated population balance models. Coalescence and break-up kernels are then

tuned to mimic bubble interactions (Sajjadi et al., 2012). This approach has been applied to stirred tank bioreactors at multiple scales from 15 mL to 70,000 L bioreactors (Haringa et al., 2017; Villiger et al., 2018). Although time-varying solutions to the flow equations are possible, the runtimes of such simulations are not practical within industrial research and design timescales.

Although the time-averaged approach described above reduces the computational complexity of the problem, it presents multiple abstractions and assumptions that are worth inspecting (Witz et al., 2016; Shu et al., 2020). Foundationally, this approach assumes that bubbles are moving through a static, time-averaged flow field with a time-independent energy dissipation landscape. In reality, the opposite is true: the time spent by an individual bubble in a bioreactor is much smaller than the time required to define a reliable mean flow field. Likewise, depending on the instantaneous state of the impeller, two bubbles at the same location but different times can experience order-of-magnitude variations in local energy dissipation rates. Since the mass transfer coefficient is a highly non-linear function of energy dissipation, a time-averaged input will necessarily distort spatiotemporal variations in mass transfer. Moreover, in most bioreactors, bubbles are discrete objects that interact directly with the fluid and other bubbles to undergo discrete break-up and pair-wise coalescence events. Recasting this discrete population set into a continuous number density is only valid if variations in the bubble population are gradual relative to the system length scale. In most mammalian stirred-tank bioreactor operating conditions, however, gas hold-up is low, and bubbles are individually identifiable. Moreover, the user-defined parameters that must be specified to build a population-balance bubble model are not always obvious.

In this work, we present a generalized framework for building time-dependent, bubble-resolved, two-phase models for investigating the real-time blending and mass transfer properties of turbulent stirred tank SU bioreactors. To begin, in Section 2 we present the relevant conservation laws that govern fluid, bubble, and species transport within an agitated tank. We emphasize transport at the level of individual bubbles and, throughout the derivation, make a direct appeal to standard transport equations. We then solve these transport equations using a combination of the lattice-Boltzmann method (for the fluid) and Newton's second law (for the bubble trajectories). The lattice-Boltzmann approach, compared to traditional finite element and finite difference approaches, provides a multiple order-of-magnitude improvement in computational speed when solving fluid transport equations (Witz et al., 2016). This trend is amplified in graphics processing unit-based computing environments, which can fully exploit the localized nature of the Boltzmann transport equation. Graphics processing units (GPUs) are computer components that were originally developed to accelerate graphics rendering and display. For some algorithms, the runtime performance realized on a desktop GPU is multiple orders-of-magnitude faster than what can be realized on a CPU cluster (Shu et al., 2020).

In Section 3, we present a systematic validation of the model by comparing predictions to experiments across multiple reactor sizes and operating conditions. To begin, we validated blend time and power dissipation predicted by the model for an un-gassed 500 L and 2000 L bioreactor against experimental data. Next, we examined fluid flow and mass transfer in a gassed 5 L benchtop bioreactor and compare the predicted mass transfer data to measurement. Sensitivities to modeling parameters are tested and discussed. Finally, we examined mass transfer in gassed 200 L, 500 L, and 2000 L bioreactors at various fill volumes, gas flow rates, and impeller speeds. Good agreement with experimental data was realized across the conditions considered here with no model reparameterization between scales or operating conditions.

In this report, we modeled gas volumes and impeller types typical of industrial mammalian bioprocesses using an approach that is purposely general. Although systems are assumed to be fully turbulent, no *a priori* assumptions have been made about the specific operating conditions. Likewise, although the bubbles are assumed to be small relative to the flow field, the physics are not tied to any specific operating scale (e.g. bench, pilot, plant). Further, when these physics are solved using algorithms tailored for GPUs, they become industrially practical. In fact, for many cases, this fully resolved implementation can generate engineering predictions faster and with fewer modeling assumptions than RANS/population balance approaches in a multi-CPU environment.

1.1. Modeling roadmap

We consider multiple bioreactor systems at multiple scales and operating conditions. In Table 1, we provide a section-by-section overview of the systems scales considered, as well as the modeling objectives, validation, and conclusions from each investigation. The aim of this step-by-step approach is to illustrate continuity between basic fluid flow and more complex mass transfer processes. Moreover, by ensuring that simpler transport processes are modeled correctly, we develop more confidence when predicting more complex phenomena.

2. Theoretical considerations

2.1. Fluid mechanics

We model fluid flow using the incompressible Navier-Stokes equations, which is solved using the lattice-Boltzmann (LB) method (Succi, 2001). The Navier-Stokes equations, which models the conservation of momentum of a fluid particle, can be written as:

$$\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \nabla \vec{V} = \vec{g} - \frac{\nabla p}{\rho} + \nabla \cdot (\nu \nabla \vec{V}), \quad (1)$$

where $\vec{V}$ is the velocity of the fluid particle at a given location at a given time t , $\vec{g}$ is the acceleration due to gravity, p is pressure, ρ is the fluid density, and ν is the fluid kinematic viscosity at a given location and time. Note that this expression is inherently transient: the left-hand side represents the acceleration of the fluid particle, while the right-hand side represents the forces due to gravity, pressure gradients, and shear stress.

The Boltzmann transport equation, which models the conservation of transport carrier probability density, can be written as (Krüger et al., 2017):

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f = \Omega(f, f), \quad (2)$$

where f represents the probability density function, $\vec{v}$ represents the phase space velocity, and Ω represents a collision operator. The Navier-Stokes equations are solved by linking f to the fluid momentum and $\Omega(f, f)$ to the fluid kinematic viscosity. When solving Eq. (2), we discretized the velocity space using a D3Q19 lattice and relax the distribution using the Bhatnagar–Gross–Krook (BGK) operator (Mohamad, 2011). Fluids interact with the tank walls via a no-slip boundary condition, as enforced via a zero-velocity bounce back method (Guo and Shu, 2013). Fluids interact with the moving impeller via the immersed boundary method, which enforces a no-slip velocity boundary condition along the impeller/shaft surfaces as they move through the fluid (Zhu et al., 2011). The top surface of the fluid is modeled as a dynamic free surface using a volume-of-fluid approach (Chiu and Lin, 2011).

Table 1
Systems considered in this report.

Section	Modeled Scale	Predicted Value	Validation	Conclusion
3.1.1	500 L and 2000 L	Blend time	Measured data from open literature	Approach correctly predicts bulk flow properties
3.1.2	500 L and 2000 L	Power input and dissipation	Measured data from open literature; conservation of energy	Approach correctly predicts power input and dissipation
3.2.2.1	5 L	Velocity field, local energy dissipation rates	Experimentally derived design correlations	The local flow field and energy dissipation rates are valid
3.2.2.2	5 L	Mass transfer coefficient	Measured data from open literature	The mass transfer coefficient is in-line with measured data
3.2.2.3	5 L	Sensitivity of output to coalescence/break-up models	Measured data from open literature	The predictions are insensitive to modeling parameters
3.3	5 L, 200 L, 500 L, 2000 L	Mass transfer coefficient	Measured data reported here	The approach works across multiple vessel scales

All models had a uniform lattice with a resolution that corresponded to 200 lattice points across the tank diameter. Although this resolution is expected to provide converged power number predictions for typical bioreactors, additional convergence tests are discussed below. The simulation time step is determined by the Courant number, Co :

$$Co = \frac{V_t}{\Delta x / \Delta t} \quad (3)$$

where V_t is the impeller tip speed, Δx is the lattice spacing, and Δt is the simulation time step. We set Co to 0.1, a value sufficient to keep lattice density fluctuations around 1%.

A large-eddy simulation (LES) turbulence closure model was applied (Pope, 2001). Within the context of LES, the unresolved (sub-grid) motions at each lattice site j are modeled by computing the local eddy viscosity, ν_{rj} :

$$\nu_{rj} = (C_s \Delta x)^2 S_j \quad (4)$$

where C_s is the Smagorinsky coefficient and S_j is the magnitude of the local strain rate tensor. When solving Eq. (1), this local eddy viscosity is calculated and added to the molecular viscosity at each time step. The Smagorinsky coefficient is set to 0.1, a value calculated from direct numerical simulation (Yu et al., 2005).

The computational models were built and solved using M-Star CFD[®], a GPU-based computational physics package tailored for chemical engineering simulations (<http://mstarcfd.com/>).

2.2. Energy input and dissipation

Fluid motion and oxygen transport within stirred tank bioreactors are governed primarily by the action of the impeller. Beyond simply homogenizing the fluid, the energy dissipation rates induced in the fluid by the impeller directly inform the bubble size distribution and local mass transfer coefficients. From the work-energy theorem, the power required to spin an impeller, P_s , is a function of the torque on the impeller, τ , and the impeller speed, N :

$$P_s = \tau(2\pi N) \quad (5)$$

This input power is often normalized by density, impeller speed, and impeller diameter, D , to calculate a non-dimensional power number, N_p :

$$N_p = \frac{P_s}{\rho N^3 D^5} \quad (6)$$

Under steady-state, fully-turbulent operating conditions, hydrofoils typically used in stirred tank bioreactors have a power number between 0.5 and 1.5 (Tuohey et al., 2018).

The local energy dissipation rate at lattice site j , ε_j , across the fluid is calculated from:

$$\varepsilon_j = \nu_{Tj} S_j^2 \quad (7)$$

where ν_T is the total (molecular plus eddy) viscosity of the fluid. The total power dissipated within the fluid, P_d , is equal to the volume integral of ε_j over all lattice sites:

$$P_d = \iiint \varepsilon_j dm \quad (8)$$

where dm is the differential mass of the fluid at lattice site j . Note that, since the lattice spacing is uniform, each lattice site has the same differential mass and a subscript is not needed.

Conservation of energy sets expectations on the relationship between P_s and P_d . During the first few seconds of operation, as the impeller accelerates the fluid from rest, the shaft power will be greater than the power dissipation. At steady state operating conditions, as the fluid kinetic energy becomes constant, P_s and P_d will converge to a similar value. As detailed in Section 3, this energy balance and the associated energy dissipation rate across the vessel directly inform mass transfer.

2.3. Bubble dynamics

2.3.1. Bubble trajectories

In this work, bubbles are modeled individually as discrete spheres assumed to be small relative to the fluid cells. Each bubble enters the system at the location of the sparge holes with a user-defined velocity, gas density, and bubble diameter sampled from a user-set diameter probability density function. Bubble trajectories are governed by Newton's second law,

$$m_i \vec{a}_i = \vec{F}_{ig} + \vec{F}_{ia} + \vec{F}_{ip} + \vec{F}_{id} \quad (9)$$

where m_i and $\vec{a}_i$ represent the mass and acceleration vector of bubble i , $\vec{F}_{ig}$ is the gravity force, $\vec{F}_{ia}$ is the added mass force, $\vec{F}_{ip}$ is the instantaneous pressure gradient force, and $\vec{F}_{id}$ is the instantaneous drag force on the bubble (Rettinger and Rüde, 2018). The trajectories of the bubbles are advanced in tandem with the fluid algorithm using the velocity Verlet algorithm (Allen and Tildesley, 2017).

In accordance with Newton's third law, particles are two-way coupled to the fluid, such that a body force at fluid lattice voxel j , $\vec{F}_{fj}$:

$$\vec{F}_{fj} = - \sum_i \vec{F}_{ia} + \vec{F}_{id} \quad (10)$$

is superimposed on the fluid lattice voxel j containing the set of bubbles $i \in j$. This fluid body force does not include the pressure gradient force on the particle, as this force is already incorporated into Eq. (1).

2.3.2. Mechanical interactions

As bubbles move through the fluid, the diameter will change due to bubble break-up events, bubble coalescence events, changes in local hydrostatic pressure, and mass transfer. Bubble break-up occurs when the local energy dissipation rate of the fluid surrounding a bubble exceeds the bubble surface energy. The diameter of the largest bubble stable against break-up, d_e , is estimated from first-principles turbulence theory (Hinze, 1955):

$$d_e \approx \frac{\sigma^{3/5}}{\rho^{3/5} \varepsilon^{2/5}} \quad (11)$$

where σ is the bubble/liquid surface tension. For a constant surface tension and fluid density, increasing the local energy dissipation rate decreases the maximum allowable bubble diameter. Mechanistically speaking, if a bubble enters a region wherein the local and instantaneous d_e defined by Eq. (11) is smaller than the current bubble diameter, a break-up event is expected. Observation and theory indicate that, at operating conditions relevant to bioreactors, break-up events are most likely to produce two daughter bubbles with an approximately 20%/80% volume split relative to the mother bubble (Sungkorn et al., 2012; Xing et al., 2015). We test this and other break-up ratios, as discussed in more detail in Section 3.

In addition to break-up, bubbles moving through the fluid will interact with each other via coalescence (collisions to form one larger bubble) and bouncing events (two bubbles bounce off each other with no change in individual bubble size). Coalescence is a mechanistically complex process that occurs in multiple stages (collision, film drainage, and film rupture) (Xing et al., 2015). A bouncing event is also complex, as it can produce temporary variations in bubble shape and mass transfer surface area. Despite this complexity, coalescence can be modeled in a tractable way by noting that in turbulent bioreactors, the coalescence and bouncing processes are fast compared to the bubble residence times and the mass transfer timescale (Krakau and Kraume, 2019). Moreover, for the modest gas hold-up volumes typically realized in SU bioreactors, the time between bubble collisions will be orders-of-magnitude longer than the time bubbles spend interacting with each other (To an order-of-magnitude). As such, the net outcome of the interaction (in terms of coalescence versus bounce) has a larger effect on mass transfer than the brief dynamics of the bubbles during interaction events.

For the low Weber number systems considered here, the net effect of the bubble pair interaction (coalesce versus bounce) can be predicted *a priori* from the bubble pair Reynolds number Re_a :

$$Re_a = \frac{U_a d_h}{\nu} \quad (12)$$

where U_a is the magnitude of the bubble approach velocity and d_h is the harmonic diameter of the bubble pair. The critical value for Re_a is determined from experiment: when Re_a exceeds 40, bubbles coalesce to create a new bubble while conserving total volume. When Re_a is less than or equal to 40, the bubble pair undergoes an elastic bounce (Boshenyatov, 2013). This rule is applied to all pair-wise bubble interactions, which are tracked in real-time and in-tandem with the fluid flow.

In addition to break-up and pair-wise coalescence, variations in the local fluid pressure along the bubble trajectory can cause an individual bubble to shrink or grow. Assuming the gas inside the bubble to be ideal, the effect of local pressure on diameter is related to Boyle's law:

$$\frac{d_i}{d_{i,r}} = \left(\frac{P_r}{P_i} \right)^{1/3} \quad (13)$$

where d_i is the current diameter of the bubble, $d_{i,r}$ is the diameter of the bubble at reference pressure P_r , and P_i is the current fluid pres-

sure around the bubble. We set the reference pressure to one atmosphere and recognize that the variation in P_i is primarily due to hydrostatics. As such, for the low gas fraction systems considered here, Eq. (13) can be recast as:

$$\frac{d_i}{d_{i,atm}} = \left(1 + \frac{\rho g h}{P_{atm}} \right)^{1/3} \quad (14)$$

where P_{atm} is the pressure in the tank headspace, $d_{i,atm}$ is the bubble diameter at atmospheric pressure, and h is the depth of the bubble below the surface. For the 2000 L bioreactors considered here, which present a fluid depth of up to 1.5 m, we expect pressure differences to change the bubble diameter by only a few percent. In the smaller vessels, these effects will be undetectable. For completeness, however, we include the effect in all simulations.

2.4. Mass transfer

On a bubble-by-bubble basis, the molar transport rate of gas from each bubble to the surrounding fluid is modeled by:

$$\dot{n}_i = H_i (S_i C_{g,i} - C_{f,i}), \quad (15)$$

where $\dot{n}_i$ is the mass transfer rate from bubble i to the surrounding fluid, H_i is the overall mass transfer coefficient of the fluid surrounding the bubble, S_i is the local dimensionless oxygen solubility of the fluid around the bubble, $C_{g,i}$ is the oxygen concentration in the bubble, and $C_{f,i}$ is the local dissolved oxygen concentration around the bubble. Mass transfer is a function of both the overall mass transfer coefficient and the local concentration difference; both must be non-zero to realize mass flux. We use trilinear interpolation to calculate the properties of the fluid at the location of the bubble.

The overall mass transfer coefficient is a function of the convective mass transfer coefficient and the bubble surface area:

$$H_i = k_{L,i} A_i, \quad (16)$$

where $k_{L,i}$ is the convective mass transfer coefficient around the bubble and A_i is the bubble surface area. For small bubbles, the local convective mass transfer coefficient is a function of the energy dissipation rate, viscosity, and Schmidt number of the surrounding fluid (Kawase and Moo-Young, 1990):

$$k_{L,i} = C (\varepsilon_i \nu_i)^a Sc_i^b, \quad (17)$$

where ε_i is the specific energy dissipation rate of the fluid surrounding the bubble, ν_i is the kinematic viscosity of the fluid surrounding the bubble, and Sc_i is the Schmidt number of the fluid surrounding the bubble. Dimensional analysis requires that b is set to $1/4$. The constant a and cofactor C are defined from first principles turbulence theory to be $-1/2$ and 0.301, respectively (Kawase et al., 1992).

Mass exchange between the bubble and the surrounding fluid causes the bubble volume to change. This volume change is related to the mass flux via the molar volume of the species undergoing transport:

$$\frac{dV_i}{dt} = -V_m \dot{n}_i \quad (18)$$

where V_i is the volume of the bubble, V_m is the gas-phase molar volume of the species undergoing transport, and $\dot{n}_i$ is the mass transfer rate defined in Eq. (18). Note that the negative sign implies that species leaving the bubble will cause it to shrink. If the gas(es) in the bubbles are ideal, the molar volume is defined from:

$$V_m = \frac{R_u T}{P_g} \quad (19)$$

where R_u is the universal gas constant, T is the temperature of the gas, and P_g is the pressure of the gas inside the bubble. Note that

the equations presented here support bidirectional transport to/from the bubbles and, although presented here for a single gas species, can be generalized to support multiple species systems.

2.5. Species transport in the fluid

Species transport within the fluid is modeled via the advection–diffusion–reaction equation:

$$\frac{dc_j}{dt} = \nabla \cdot (D \nabla c_j) + \nabla \cdot (\vec{V}_j c_j) + \frac{\dot{n}_j}{V_j} + R_j \quad (20)$$

where c_j is the species concentration in fluid lattice voxel j , D is the gas diffusion coefficient, $\vec{V}_j$ is the fluid velocity vector, V_j is the volume of the fluid lattice voxel, $\dot{n}_j$ is the species transfer rate between the bubbles and the fluid, and R_j is the local species reaction rate. We use the lattice-Boltzmann particle velocities to solve Eq. (20) at each timestep for each fluid voxel (Küng et al., 2020). A van Leer flux limiter is applied to control numerical diffusion (Van Leer, 1979). From Section 2.3, the source term in Eq. (20) can be recast as:

$$\dot{n}_j = \sum_{i \in j} k_i A_i (S_j c_{g,i} - c_{f,j}) \quad (21)$$

Note that i denotes the properties calculated for individual bubbles, while j represents properties of individual fluid voxels. The sum is over the set of all bubbles with centers of mass contained within voxel j .

It should be noted that the transport physics are modeled here at the level of individual bubbles and fluid voxels. If the system is assumed to be well-stirred (to achieve a uniform dissolved oxygen concentration c_l) and the gas concentration in the bubbles is assumed to be a constant, Eq. (20) can be simplified into:

$$\frac{dc_l}{dt} = \frac{k_l A (S c_b - c_l)}{V_l} = k_l a (S c_b - c_l), \quad (22)$$

where c_b is the gas concentration in the bubbles. In the middle expression, k_l is the total effective convective mass transfer coefficient of all the bubbles, A is the total surface area of all the bubbles and V_l is the volume of the liquid. In the right-hand expression, a is the specific surface area of the bubbles in the vessel. The coefficient $k_l a$, which has units of reciprocal time, is called the volumetric mass transfer coefficient. This coefficient is typically determined experimentally by fitting the measured time-evolution of the dissolved oxygen concentration in the fluid to the right-hand side of Eq. (22).

3. Validation

3.1. Single phase blend time and power draw in 500 L and 2000 L SU bioreactors

3.1.1. Blend time

To begin validation, we compare the predicted to measured blend times for a 500 L and 2000 L SU bioreactors operating at various fill levels and impeller speeds. These reactors, as reproduced in Fig. 1, are geometrically patterned after the design of the GE Xcellerex Stirred Tank Bioreactor series (GE Healthcare KA4969250918AN; GE Healthcare 29243481AA). The impeller in both systems is bottom-mounted but off-center, such that the impeller rotation vector is tilted 15 degrees relative to the tank centerline. The vessel diameters are 0.76 and 1.2 m, respectively. The impeller in the 500 L vessel is a three-bladed hydrofoil with a diameter of 0.26 m. A similar hydrofoil is used in the 2000 L vessel, although it has four blades and a diameter of 0.43 m.

GE characterized the blend time of these bioreactors using an acid/base neutralization experiment (GE Healthcare KA4969250918AN; GE Healthcare 29243481AA). Within this pro-

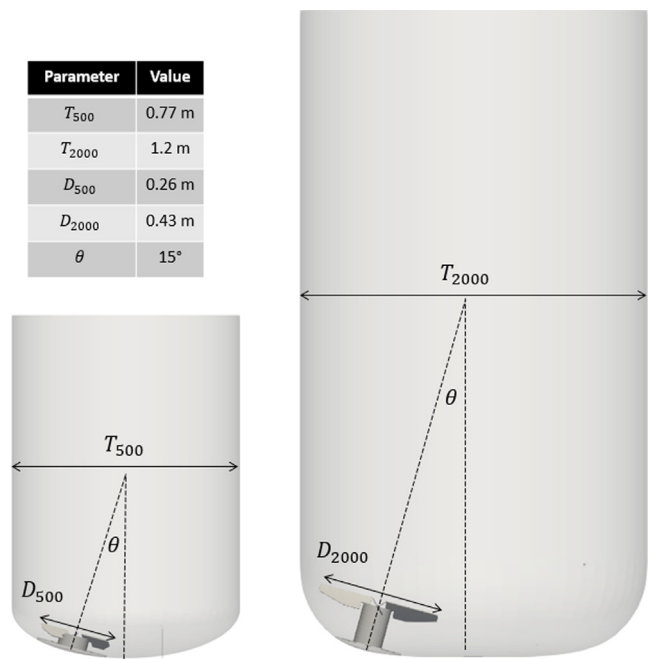


Fig. 1. Relevant dimensions of the 500 L and 2000 L single-use bioreactors. The tank designs are geometrically similar, although the impellers differ between the two systems. In the 500 L vessel, the impeller is three-bladed and adjacent to the bottom wall. In the 2000 L vessel, the impeller is four-bladed and offset from the bottom wall.

cess, a small quantity of acid (or base) was added to the top of the fluid and the local fluid pH was tracked using an array of probes across the tank. The time required for the last probe to stabilize to within 5% of its steady state pH value is defined as the system blend time. We simulated this experiment, as shown in Fig. 2, using a similar approach. We add 0.5 mol of weightless dye to the top of the fluid. The time required for the global coefficient of variation (COV) across all fluid voxels to stabilize below 5% was taken as the blend time. Because the COV is calculated across the entire flow domain, the approach is guaranteed to incorporate the last regions of the fluid to blend.

In simulations involving the 500 L tank, the fluid was agitated for 30 s prior to dye addition to establish a steady-state flow field, during which the total kinetic energy of the fluid was constant. This initialization period was increased to 60 s for the 2000 L vessel simulations. The liquid density and viscosity were set to 1000 kg/m³ and 10^{−6} m²/s. The dye was assigned a diffusion coefficient of 10^{−9} m²/s, which is typical of gases within liquids (Lide, 2004). A van Leer flux limiter was applied to control numerical diffusion. The simulation resolution in both cases was set to 200 points across the tank diameter. These resolutions were selected to achieve converged power numbers, as discussed below.

In Fig. 3, we present a comparison between the model predicted (“Predicted”) and experimentally measured (“Measured”) blend times across a variety of fill levels and impeller speeds. To contextualize this data, we superimpose on these results the correlation estimates (“Correlation”) from an experimentally derived blend time relationship by Grenville (1992):

$$\theta_{95} = \frac{5.2 T^{3/2} H^{1/2}}{N D^2 N_p^{1/3}}, \quad (23)$$

where θ_{95} is the 95% blend time and H is the fluid height. The agreement between the three approaches is good. Most values are within 20%, which agrees with the confidence interval of the blend time correlation. A few exceptions are worth noting: For Case 16, the

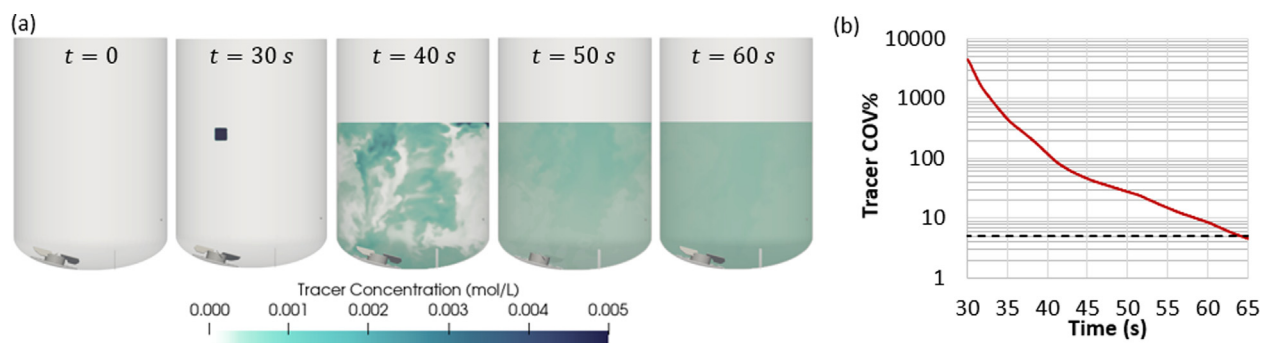


Fig. 2. (a) Time evolution of dye concentration across the fluid for the 500 L vessel with a 300 L operating volume and an impeller speed of 115 RPM. (b) Time evolution of the global percent coefficient of variation (COV) of the scalar field, which was added at 30 s. The predicted blend time here is 32.3 s, which is the time at which the COV crosses below 5%.

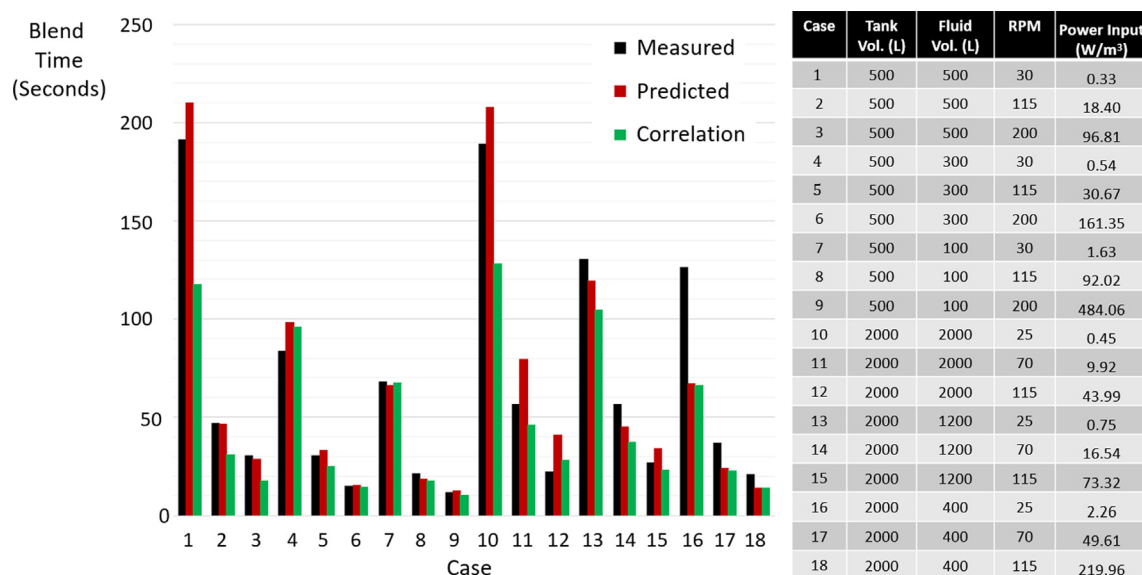


Fig. 3. Predicted versus measured blend time for the 500 L and 2000 L SU bioreactors at various fill levels and impellers speeds.

experimental measurement exceeds the simulation predictions and design correlation by a factor of 2x. Trends from the measured data for Cases 10 and 13, however, suggest that this difference may be related to measurement uncertainty. Also, when both the 500 L and 2000 L are operating at their maximum fill level and minimum impeller speed, experiment and simulation both suggest a blend time between 190 and 210 s. Eq. (23) predicts a much shorter blend time. This deviation is not surprising, since Eq. (23) was developed for vessels with fluid levels close to the tank diameter. For most of the operating conditions, the agreement between the three approaches substantiate the measured data and corroborate the validity of the mass/momentum modeling approach.

3.1.2. Power draw

In Fig. 4, we compare the predicted shaft power, P_s , predicted total power dissipation, P_d , and measured shaft power for Case 2. The predicted shaft power was calculated using Eq. (5), while the total power dissipation was calculated using Eq. (8). At early times, as discussed in Section 2.2, the shaft power exceeds the energy dissipation as fluid accelerates from rest. After about 30 s of agitation, upon which the system reaches a steady state operating condition, these two values converge to average value of 8 W, with fluctuations on the order of 0.8 W ($\pm 10\%$). The measured steady-state turbulent power number for the 500 L system is 1.1 (GE Healthcare

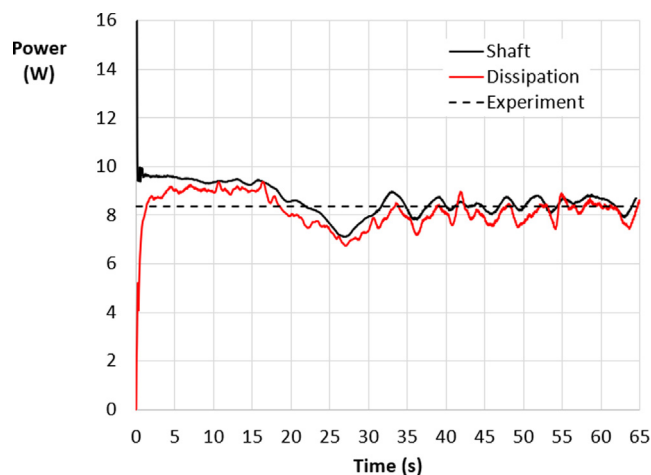


Fig. 4. Time-evolution of the shaft power and power dissipation for Case 2. Superimposed on this data is the time-average shaft power corresponding to the measured power number of 1.1.

KA4969250918AN), which corresponds to 8.05 W of power input at this impeller speed. Both P_s and P_d were within 5% of this measured value. Similar agreement between P_s and P_d was predicted

for the 2000 L system, although the predicted impeller power number for the 2000 L system was 0.85. The difference in power number between the two scales, which arises from differences in the impeller topology and position, is consistent with measurement (GE Healthcare KA4969250918AN).

It is important to recognize that, within the context of the solver, the shaft power and power dissipation calculations are fully independent. The solver does not explicitly enforce the conservation of energy; conservation is instead satisfied by the flow field. The agreement between P_s and P_d therefore reaffirms the self-consistency of the solver, while the agreement with the measured value suggests consistency with experiment. This consistency is important, as accurately predicting energy dissipation is a key step in predicting bubble size and mass transfer.

Increasing the resolution beyond 200 points across the tank presents no systematic change in the predicted blend time or power predictions outside the 10% uncertainty presented above. Likewise, decreasing the Courant number below 0.05 has no meaningful effect on the blend time and power number. Decreasing the resolution below 200 lattice points across the tank diameter, however, leads to overpredictions of the impeller power number and underpredictions of blend-time. Likewise, increasing the Courant number to values above 0.1 leads to unacceptable variations in lattice density. Both these sensitivities were expected *a priori* and motivated the initial resolution.

3.2. Predicting mass transfer in 5 L bioreactors

3.2.1. Simulation Set-up

Having validated the single-phase blend time and power dissipation predictions, we next extended the model to include bubbles and bubble/fluid mass transfer. The first system we study, as presented in Fig. 5, was a 5 L benchtop system patterned after the Applikon Autoclavable Bioreactor (Zakrzewski, 2020). The reactor is agitated by a single, wide-blade pitch blade turbine that was centered within the vessel and positioned directly above the gas line. The gas line had seven, 0.7 mm-diameter perforations that act as a sparge. A dissolved oxygen probe was placed near the vessel wall.

For all runs at this scale, the initial liquid volume was set to 3.9 L and the system was gassed with pure oxygen at standard temperature and pressure. The liquid density and viscosity were set to

1000 kg/m³ and 10⁻⁶ m²/s. The dimensionless solubility of dissolved oxygen in the liquid was set to 0.032 (Sander, 2015). No dissolved oxygen was initially present in the liquid prior to gassing. The model accounted for the effects of liquid volume during bubble injection and removal. Accordingly, adding gas bubbles to the system causes the free surface level to rise.

Gas bubbles entered the system at the location of the sparge holes. The initial gas bubble size was set to 3 mm, a value motivated by correlation (Rathore et al., 2012) and confirmed by visual inspection across the range of operating conditions investigated here (Zakrzewski, 2020). The density of the oxygen bubbles entering the system was set to 1.2 kg/m³ with a molar volume of 22.4 L/mol. During bubble break-up events, the volume fraction of the first daughter was sampled from a triangle distribution function with a lower limit of 0, an upper limit of 0.5, and a mode of 0.2. The volume fraction of the second daughter was the complement to the first, which ensured volume conservation over the break-up event. Gas bubbles leaving the system through the free surface cause the free surface level to fall. During steady state operating conditions, the position of the fluid level relative to the initial un-gassed position is correlated to the gas hold-up volume.

The simulation resolution was set to 200 points across the tank diameter, which corresponds to an 875 μ m grid spacing. The Courant number was set to 0.1, which corresponds to timesteps ranging from 70 μ s to 950 μ s depending on the impeller speed.

3.2.2. Model output

3.2.2.1. Velocity, energy dissipation, and bubbles. In Fig. 6, we present a snapshot of this model operating with an impeller speed of 400 RPM and a gas flow rate of 24 standard liters per hour. A plane crossing through the center of the tank shows the instantaneous fluid speed after 30 s of operation. As expected, the fluid respects the no-slip boundary condition: adjacent to the tip of the impeller, the fluid velocity is linked to the impeller tip speed (1.2 m/s in this case). At the tank wall, the velocity is zero. These results are expected, given that we are modeling a fully transient system. Between these two extremes, the velocity varies as a function of position relative to the impeller. Surface vortexing at the free surface appears negligible. After about 10 s of operation, the bubble count becomes steady. The gas hold-up, as measured by the change in the free surface position due to gassing, was less than 1%.

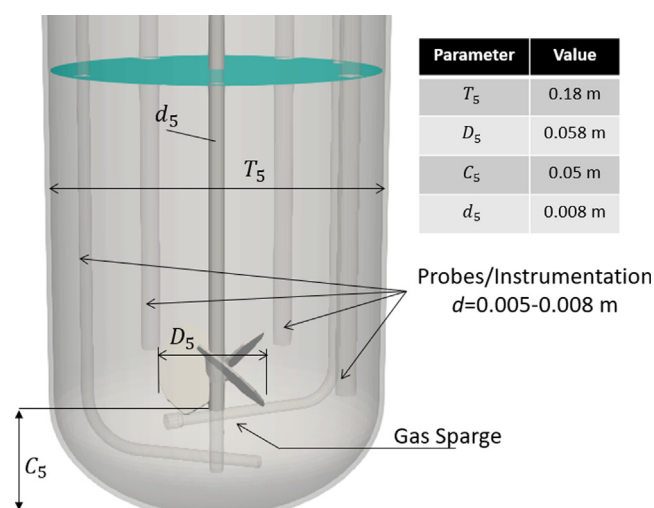


Fig. 5. Key parameters of the 5 L benchtop bioreactor. Initial position of the free surface corresponds to a fluid volume of 3.9 L. The sparge holes were on the bottom of the pipe. Multiple probe and instrumentation objects, with a range of diameters noted in the figure, act as fluid baffles.

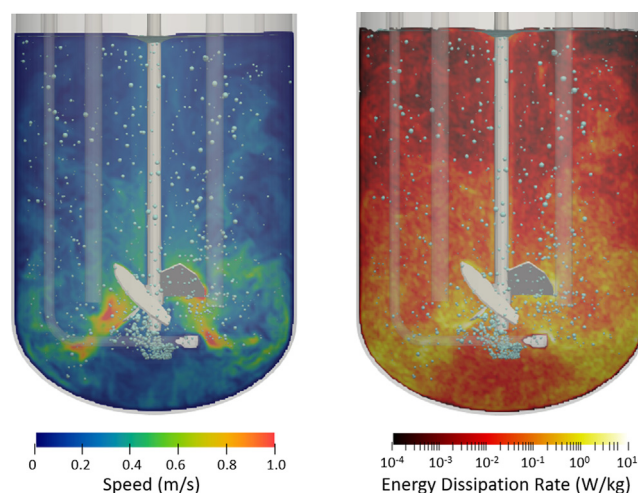


Fig. 6. Left: Instantaneous snapshot of the fluid velocity (magnitude) in a plane cut through the tank center. Flow near the sparge is dominated by the impeller trailing vortex. The bubble concentration is highest in the immediate vicinity of the sparge. Right: Spatial variations in the local energy dissipation rate across the tank. Dissipation rates near the impeller are multiple orders-of-magnitude greater than those in the tank far-field.

We superimpose on this velocity field the instantaneous, three-dimensional bubble field realized in the vessel. The bubble number density and bubble surface area are highest in the immediate vicinity of the sparge. Outside this injection region, the bubbles are homogeneously distributed across the tank by the action of the impeller. Note that the velocity field surrounding the sparge is dominated by the trailing vortex of the impeller. Likewise, for both the gassed and un-gassed systems, the predicted power number of the impeller was 1.1. This value is consistent with published data (Zanghi et al., 2017). Thus, although the bubbles were two-way coupled to the fluid, gas flow at this operating condition has no material effect on fluid flow in the trailing vortex.

In Fig. 6 (right) we present a snapshot of ε_j , as calculated from Eq. (7). The data here is presented on a log scale, which helps illustrate the stark spatial variations in ε_j across the fluid. Within the trailing vortex, the energy dissipation rate is on the order of 1–10 W/kg. Away from the impeller, the energy dissipation rate decreases to values between 0.001 and 0.01 W/kg. As energy dissipation has an important effect on mass transfer, we consider two approaches for verifying this prediction. First, using Eq. (8), we calculate the total energy dissipation P_d to be 4.8 W. This value compares well against the predicted shaft power of 4.7 W, as calculated from Eq. (5). This agreement indicates that the solution is self-consistent and conserves total energy. Second, we can corroborate that the predicted energy dissipation in the trailing vortex agrees with measured data for similar systems. For Rushton, Pitch Blade, and Hydrofoil impellers, the measured energy dissipation rate in the trailing vortex, ε_m , can be approximated from (Grenville et al., 2017):

$$\varepsilon_m = \frac{N_p N^3 D^3}{4w_p}, \quad (24)$$

where w_p is the projected blade width (0.025 m for this system). At these operating conditions ε_m is estimated to be 3 W/kg in the trailing vortex—a value in-line with the results presented in Fig. 6. The self-consistency of the solver and the agreement with measured data suggest that the predicted energy dissipation rates within the fluid are valid.

3.2.2.2. Volumetric mass transfer coefficient (k_1a). In Fig. 7, we present a snapshot of the local time-averaged volumetric mass transfer coefficient, $(k_1a)_j$, across the three-dimensional fluid domain. This value is calculated from:

$$(k_1a)_j = \frac{1}{\tau} \int_0^\tau \frac{k_{1j}(t)A_{Tj}(t)}{V_j} dt \quad (25)$$

where k_{1j} is the convective mass transfer coefficient in voxel j at time t , $A_j(t)$ is the total bubble surface area in fluid voxel j at time t , V_j is the volume of the lattice voxel, and τ is the integration time window. Note that, although $k_{1j}(t)$ is non-zero for all fluid voxels during steady state operation, $A_{Tj}(t)$ may be instantaneously zero if the voxel contains no bubbles. As such, time-averaging the data presents a more continuous representation of the field over the timescales relevant to mass transfer. To generate this image, data was time-averaged every timestep for 30 s following one minute of agitation. The required length of this sampling interval is informed by the timescales governing fluid flow and bubble transport. The 30 s averaging interval considered here captures hundreds of impeller rotations and the trajectories of several hundred thousand individual bubbles. We find that this sampling is sufficient to produce a converged $(k_1a)_j$ profile.

The transfer coefficient varies by multiple orders-of-magnitude across the vessel. This behavior is a direct consequence of spatial variations in ε and the local bubble size/concentration. Near

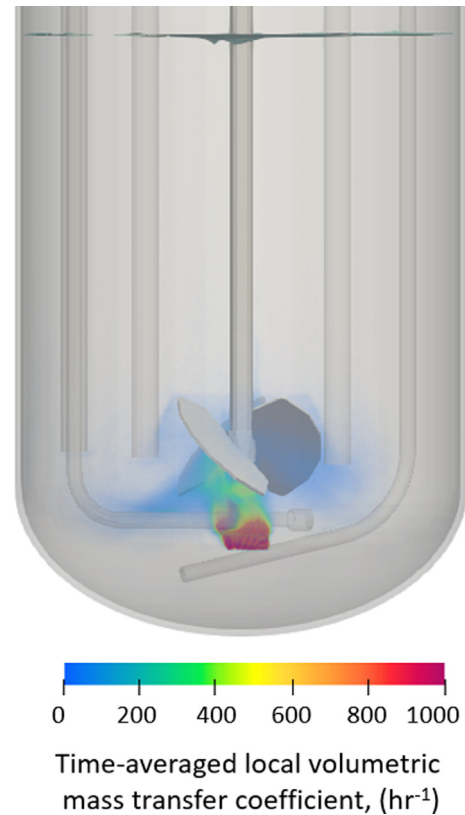


Fig. 7. Four-dimensional representation of the local volumetric mass transfer coefficient, $(k_1a)_j$. No coloring is shown in regions where the local magnitude of $(k_1a)_j$ is less than 0.1 hr^{-1} . The volume-averaged value, $\bar{k}_1a$, is computed from Eq. (26) to be 4.82 hr^{-1} .

the sparge and impeller, owing to a combination of a high bubble surface area and energy dissipation rate, $(k_1a)_j$ is on the order of 100–1000 hr^{-1} . In the tank far-field, due to lower specific bubble surface areas and energy dissipation rates, $(k_1a)_j$ is orders-of-magnitude smaller than what is realized near the impeller. In fact, in over 90% of the fluid volume, the local volumetric mass transfer coefficient is less than 1 hr^{-1} .

The overall volumetric mass transfer coefficient of the system, k_1a , can be calculated by averaging the local value over the entire system volume:

$$k_1a = \frac{1}{V_t} \iiint (k_1a)_j dV_i \quad (26)$$

Following this approach, we calculate k_1a to be 4.8 hr^{-1} . This value agrees with the measured k_1a value of $5.1 \pm 0.2 \text{ hr}^{-1}$ at this operating condition (Zakrzewski, 2020).

In Fig. 8, we present a snapshot of the instantaneous dissolved oxygen concentration profile across the system after 30 s of operation. The results are normalized by the saturated dissolved oxygen concentration. Despite strong variations in the oxygen transfer rate to the liquid, the instantaneous dissolved oxygen concentration across the fluid has a relative standard deviation of less than 5%. To understand this result, recognize that blend time in this system is characterized in seconds whereas the mass transfer timescale is characterized in hours. Although most oxygen is transferred to the liquid in the immediate vicinity of the sparge, the mixing process relatively quickly homogenizes this oxygen-rich water across the bulk of the tank. That is, at this operating condition, the fluid is well-stirred relative to the mass transfer process.



Fig. 8. Instantaneous snapshot of the dissolved oxygen concentration across the fluid after 30 s of agitation. Value normalized by the saturation limit 0.00145 mol/L. Due to differences between the blend time and mass transfer timescales, the concentration is uniform relative to the variations in the underlying mass transfer coefficient ($k_L a$).

This behavior conveys an important point: when characterizing systems with no or slow oxygen consumption (relative to the blend time), differences between the blending timescales and mass transfer timescales can mask strong spatial variations in the oxygen transfer rate. Within small/well-stirred systems, the notion of a tank-averaged $k_L a$ value is appropriate. In larger vessels or if reactions are present, spatial variations in $k_L a$ can lead to spatial variations in oxygen concentration (Lewis et al., 2016). This tank-averaged $k_L a$ value becomes a less reliable metric for predicting system performance, and $(k_L a)_j$ should be considered. This point is discussed further in Section 3.3.

In Fig. 9, we present the time-evolution of the dissolved oxygen concentration at the probe, normalized by the saturated oxygen concentration. We calculated $k_L a$ by fitting this data to Eq. (22), as shown in the figure. From this fit, $k_L a$ is calculated to be 5.0 hr^{-1} , a result that agrees with the value of 4.8 hr^{-1} calculated using Eq. (26) and the measured value of $5.1 \pm 0.2 \text{ hr}^{-1}$ (Zakrzewski, 2020). The agreement between two independent simulation predictions [probe-based calculation of Eq. (22) and the volume average calculation of Eq. (26)] indicates that mass is conserved in the system and the species transport algorithms are self-consistent.

3.2.2.3. Model sensitivities. The predicted gas transfer properties are related to the bubble surface area, which is a function of the user-defined bubble break-up mechanics. To understand potential sensitivities to this parameter, in Fig. 10, we present the predicted $k_L a$ as a function of the specified bubble break-up kernel. In addition to the triangle distribution applied above, separate simulations with break-up into (i) equal volume daughters and (ii) daughters with a random volume ratio were tested. A system with no break-up

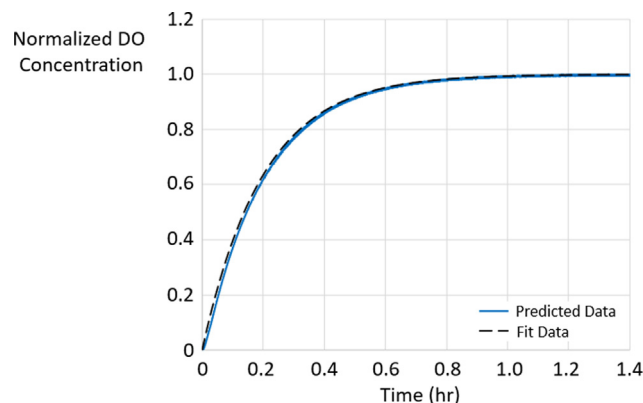
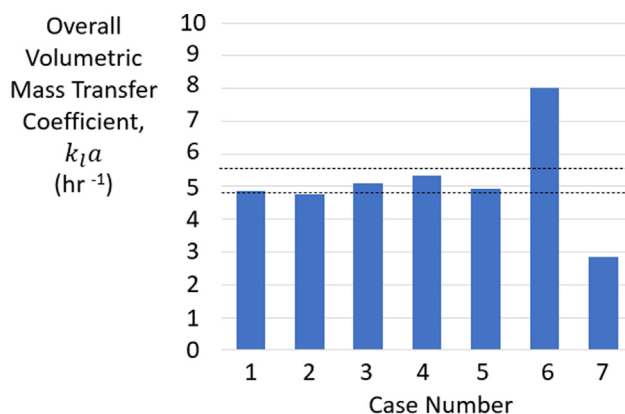


Fig. 9. Time-evolution of the dissolved oxygen concentration at the probe. The concentration is normalized by the saturation limit of 0.00145 mol/L. By fitting the data to Eq. (22), the overall volumetric mass transfer coefficient is calculated as 5.0 hr^{-1} .



Case	Initial Bubble Dia. (mm)	Break-up	Coalescence	$k_L a \text{ (hr}^{-1}\text{)}$
1	3	Triangle	Active	4.86
2	3	Half	Active	4.75
3	3	Random	Active	5.11
4	1-3	Triangle	Active	5.33
5	5	Triangle	Active	4.93
6	3	Triangle	Inactive	8.03
7	3	Inactive	Inactive	2.84

Fig. 10. Effect of break-up kernel, coalescence, and initial bubble size on the predicted overall volumetric mass transfer coefficient, $k_L a$. Range of measured data for this system presented as dashed lines.

was also tested. We superimpose on these predictions the corresponding minimum and maximum measured values at this operating condition. The $k_L a$ predicted using each break-up kernel were consistently within the range of measured values, which are indicated by the dashed horizontal lines in the figure. This insensitivity to the break-up kernel does not imply an altogether insensitivity to break-up or changes in bubble size. If break-up is not modeled, $k_L a$ is underpredicted 2x relative to experiment. If coalescence is ignored, $k_L a$ is overpredicted by a factor of 1.5x.

We also present the predicted $k_L a$ as a function of the initial bubble size (with break-up and coalescence both active). In addition to the 3 mm initial bubble diameter applied above, separate simulations with injection diameters of (i) 0.7 mm and (ii) 5 mm were tested. A system wherein the bubble diameters were randomly sampled at runtime from a uniform distribution between 1 mm and 3 mm was also tested. The predicted $k_L a$ using each ini-

tial bubble diameter were, again, consistently within the range of measured values.

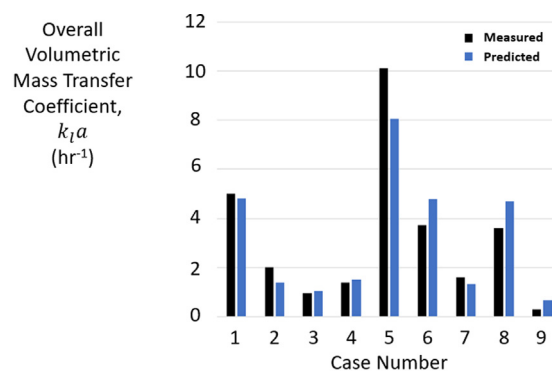
The relative insensitivity of the simulation predictions to the initial bubble size and breakup kernel is a function of the system topology and operating condition. At this impeller speed/gas injection rate, as discussed above, the flow field in the tank is dominated by the impeller; the gas injection has no material effect on power draw and the fluid velocity field is consistent with that of a single-phase system. Likewise, in both experiment and simulation, the gas bubbles emerge from the sparge as discrete objects that rise directly towards the impeller. Under these conditions, the fluid motion in the trailing vortex will establish and enforce a bubble size distribution—regardless of the nuances of the break-up kernel or initial condition. At higher gas flow rate conditions, as the impeller begins to flood and the trailing vortex deforms in response to the gas flow (Hudcova et al., 1989), the predicted results are expected to become increasingly sensitive to the break-up kernel. Likewise, if the gas flow rate is sufficient to induce jetting at the sparge outlets (Syeda and Ansery, 2014), the predictions will become increasingly sensitive to initial bubble size.

3.3. Testing other operating conditions and scales

We next used the model to investigate other operating conditions and system scales. From this work, we sought to assess the generality of the mass transfer framework and modeling approach presented in Section 2. In addition to the 5 L benchtop system introduced in Section 3.2, we also considered mass transfer in 200 L, 500 L, and 2000 L SU bioreactors. The 500 L and 2000 L systems are the same systems used in Section 3.1 to validate blend time and power dissipation. The 200 L system is geometrically identical to the 500 L vessel, but with a tank diameter T_{200} of 0.57 m and an impeller diameter D_{200} of 0.22 m (GE Healthcare KA686051118DF).

Gas bubbles entered the 200 L, 500 L and 2000 L bioreactors through 1 mm holes positioned on the wall immediately below the impeller. Consistent with the physical systems, the 200 L and 500 L vessel models had 20 holes, while the 2000 L vessel had 40 holes. We only considered gas flow rates in the non-jetting regime, such that modeling individual bubbles issuing from the individual sparge holes is appropriate. The initial bubble diameter between cases ranged from 1 mm to 5 mm, which was representative of observation and literature correlations (Rathore et al., 2012). We found, within this range of values, the predicted mean bubble size and $k_L a$ values were largely insensitive to the initial bubble size. The triangle break-up kernel was applied to each model. Coalescence was active in each simulation. All systems used a resolution of 200 lattice points across the tank diameter and a Courant number of 0.05.

In Fig. 11, we compare the predicted $k_L a$ for each system to experimentally determined values at the listed working volumes, impeller speeds and gas flow rates. Experimental design and results are presented in Supplemental Information. The results from Section 3.1 are included in the data as Case 1. Across the range of operating conditions here, the predicted values agreed with the measured conditions. No adjustments to the mass transfer physics were made between simulations. The consistency of the agreement across these scales reaffirms the generality of the modeling approach. The measured gasified power number for these systems is consistent with the un-gassed power number (GE Healthcare KA4969250918AN; GE Healthcare 29243481AA). This finding confirms that the flow field near the impeller is largely unaffected by the gas flow. As discussed in Section 3, this behavior suggests insensitivity to the selected break-up kernel.



Case	System	Fluid Volume (L)	Gas Rate (SLPM)	Impeller Speed (RPM)	Power Input (W/m ³)
1	Bench 5 L	3.9	0.4	400	54.900
2	Bench 5 L	3.9	0.4	30	0.023
3	Bench 5 L	3.9	0.1	300	23.100
4	SU 200 L	200	5	75	5.540
5	SU 500 L	250	25	125	47.300
6	SU 500 L	375	13	100	16.100
7	SU 500 L	500	5	75	5.110
8	SU 2000 L	1500	45	90	28.100
9	SU 2000 L	2000	5	75	12.200

Fig. 11. Predicted $k_L a$ across a range of system volumes and operating conditions.

It is important to recognize that, although two systems at two different scales may have similar measured/predicted $k_L a$, the actual processing environment within each tank can vary significantly. To illustrate this point, in Fig. 12 we present simulation output from Case 8: a 2000 L bioreactor operating at a fill volume of 1500 L. The measured/predicted $k_L a$ for this system is similar to those measured/predicted for Case 1, which is a 5 L bench-scale system. From this parameter alone, one might incorrectly conclude that biomass within the two reactors experience a similar environment. Mixing in the larger system, however, is ineffective relative to the smaller system. As such, bubbles in the larger system are not homogeneously distributed across the fluid; most entering the vessel exit the trailing vortex and rise directly to the free surface. This behavior produces an asymmetric mass transfer coefficient across the vessel. The asymmetric mass transfer coefficient leads to larger spatial variations in the local dissolved oxygen concentration. In this larger tank, over the bulk of the oxygen transfer process, the dissolved oxygen concentration COV is on the order of 10–30%. This value is much larger than the 1–10% dissolved oxygen COV predicted for the benchtop system. In the absence of oxygen consumption, these gradients will eventually disappear as the fluid becomes oxygen saturated. During production cultures, however, cellular oxygen consumption will exacerbate these spatial variations in concentration. This performance difference between scales is not obvious from the overall $k_L a$ measurement alone.

4. Computational considerations

On a desktop workstation with four Nvidia Tesla V100 GPUs, simulations advanced at 1.5 billion lattice updates per second. Thus, for the 2 to 16 million lattice point simulations considered here, each second of simulation time required about 0.5–1 min of wall time. Simulation runtimes vary linearly with the number of CUDA cores allocated to the simulation; running on eight V100 GPUs enabled simulations to run 2x faster (3 billion lattice updates per second). Running on one V100 GPU typically realized 0.4–0.5 billion lattice updates per second. We note that, even on one GPU, it is practical to simulate tens of minutes of process dynamics per day of computer time.

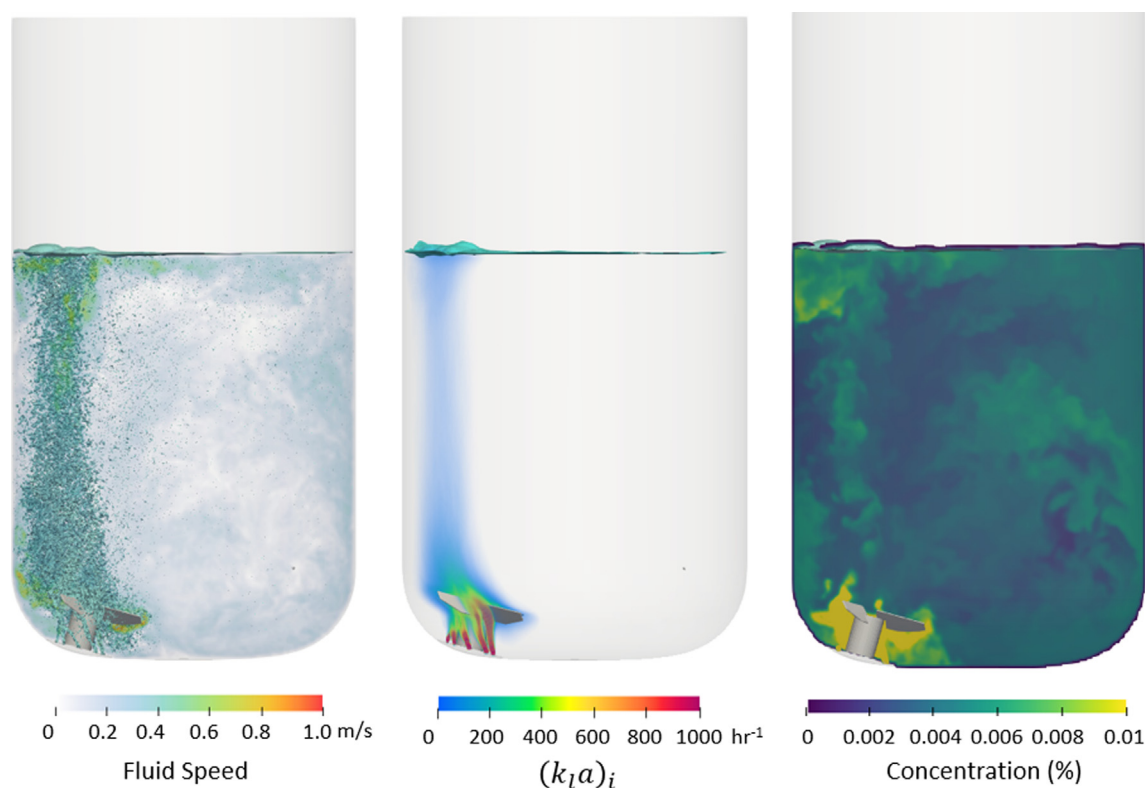


Fig. 12. Fluid velocity, local volumetric mass transfer coefficient $(k_L a)_i$, and oxygen concentration in the 2000 L vessel with a 1500 L fill volume after 60 s of operation. Concentration normalized by the fully saturated value of 0.00145 mol/L.

In principle, the two-phase approach presented here is independent of the system volume. In practice, however, the algorithm is limited by (i) the maximum number of discrete bubbles that can be stored in GPU memory and (ii) the efficiency of the bubble collision algorithm. Roughly speaking, each gigabyte of GPU RAM can hold about 1,000,000 particles. And, on a multi-GPU desktop workstation, we find that simulations containing up to 10^6 – 10^7 individual bubbles have a runtime performance similar to single-phase blend time simulations. That is, within this range, the additional computational time required to compute bubble dynamics and interactions is small relative to the fluid and species solvers. Most SU bioreactors at typical operating conditions contain fewer than ten million bubbles. As such, these systems will not approach the capacity limit or performance of desktop GPUs. When simulating large-scale ($>10\text{ m}^3$) systems with high large gas hold-up ($>10\%$) and small average bubble diameters (less than 1 mm), however, a GPU cluster is required.

5. Conclusion and next steps

In this paper, we have derived, implemented, and validated a mechanistic, multi-phase model for predicting species transport and $k_L a$ within gassed, stirred-tank bioreactors over a large range of operating volumes and conditions. This approach was designed deliberately to be general with respect to tank configuration, impeller type, and impeller position. We began in Section 2 by outlining the key physics driving transport in these systems, in terms of the conservation of momentum and species. Particular attention was given to the bubbles, which are modeled here as discrete objects undergoing explicit break-up and pair-wise coalescence events. Next, in Section 3, we validated the predicted blend times for 500 L and 2000 L vessels over a range of operating conditions against experimental data. We

also confirmed that power dissipation in the fluid—a key parameter informing mass transfer—was modeled properly in the system and consistent with experiment.

We continued in Section 3 by investigating mass transfer in a 5 L bench-scale bioreactor. The overall volumetric mass transfer coefficient was consistent with the measured values. Large variations in energy dissipation rate and bubble concentration were predicted across the fluid, however, which lead to large variations in the local mass transfer coefficient within the vessel. The model was found to be largely insensitive to the specifics of the bubble break-up kernel and initial bubble size. We then tested the model across a range of operating conditions for system volumes ranging from 5 L to 2000 L. Good agreement was realized across the range of systems considered here, with no reparameterization or model adjustment between scales. We observe that, for the non-reactive systems considered here, oxygen gradients across the system will disappear as the fluid reaches a saturated limit.

The approach presented here can be extended to handle multi-species transport, to include combined oxygen transport with carbon dioxide scrubbing. Similarly, it is straightforward to incorporate reactions into the conservation of species equation and solve them in real-time. Used in this way, in addition to species exchange, the approach can be further generalized to model nutrient consumption and drug product growth. Because the fluid, species, and bubble transport physics are solved in tandem and in real-time, this modelling approach can be used to study time-evolving, multi-physics production rates. We recognize that modifications to this approach may be necessary for systems with larger bubble populations, if the number of tracked bubbles exceeds available GPU memory. One approach for addressing this limitation is to use a coarse-grained particle method, wherein several real bubbles are lumped into a single tracked parcel (Nasato et al., 2015).

The success of the model across these scales and operating conditions is linked to the fundamentality of the physics evoked during development. When deriving the model, we made direct appeals to time-accurate, three-dimensional mass and momentum transport equations for the fluid and bubbles. Structured in this way, the transport equations become independent of the specifics of the bioreactor scale. The computational requirements of this modeling approach are, admittedly, beyond the capability of most finite element and finite difference-based solvers. When these physics are solved using algorithms tailored for GPUs, however, they become industrially practical. In fact, for many cases, this transient, fully resolved implementation can generate engineering predictions faster and with fewer modeling assumptions than steady-state RANS/population balance approaches in a multi-CPU environment.

CRedit authorship contribution statement

John A. Thomas: Conceptualization, Methodology, Data curation, Validation, Writing - original draft. **Xiaoming Liu:** Conceptualization, Methodology, Data curation, Validation, Writing - original draft. **Brian DeVincentis:** Conceptualization, Methodology, Software, Writing - review & editing. **Helen Hua:** Conceptualization, Methodology, Validation, Writing - review & editing. **Grace Yao:** Conceptualization, Methodology, Validation, Writing - review & editing. **Michael C. Borys:** Conceptualization, Methodology, Supervision, Writing - original draft, Writing - review & editing, Resources. **Kathryn Aron:** Supervision, Project administration, Funding acquisition, Writing - review & editing, Resources. **Girish Pendse:** Supervision, Project administration, Funding acquisition, Writing - review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Allen, Michael P., Tildesley, Dominic J., 2017. *Computer Simulation of Liquids*. Oxford University Press.
- Amer, Momen, Feng, Yu, Ramsey, Joshua D., 2019. Using CFD simulations and statistical analysis to correlate oxygen mass transfer coefficient to both geometrical parameters and operating conditions in a stirred-tank bioreactor. *Biotechnol. Prog.* 35 (3), e2785.
- Bhole, M.R., Joshi, J.B., Ramkrishna, D., 2008. CFD simulation of bubble columns incorporating population balance modeling. *Chem. Eng. Sci.* 63 (8), 2267–2282.
- Boshenaytov, Boris, 2013. Laws of bubble coalescence and their modeling. *J. Magnetohydrodynamics Plasma Res.* 18 (4), 311.
- Chiu, Pao-Hsiung, Lin, Yan-Ting, 2011. A conservative phase field method for solving incompressible two-phase flows. *J. Comput. Phys.* 230 (1), 185–204.
- Delafosse, Angélique, Calvo, Sébastien, Collignon, Marie-Laure, Toye, Dominique, 2018. Comparison of hydrodynamics in standard stainless steel and single-use bioreactors by means of an Euler-Lagrange approach. *Chem. Eng. Sci.* 188, 52–64.
- Dhanasekharan, Kumar M., Sanyal, Jay, Jain, Anupam, Haidari, Ahmad, 2005. A generalized approach to model oxygen transfer in bioreactors using population balances and computational fluid dynamics. *Chem. Eng. Sci.* 60 (1), 213–218.
- Doran, P.M., 1995. *Bioprocess Engineering Principles*. Elsevier.
- Dreher, Thomas, Husemann, Ute, Adams, Thorsten, de Wilde, Davy, Greller, Gerhard, 2014. Design space definition for a stirred single-use bioreactor family from 50 to 2000 L scale. *Eng. Life Sci.* 14 (3), 304–310.
- Elqotbi, Mohammed, Vlaev, S.D., Montastruc, Ludovic, Nikov, Iordan, 2013. CFD modelling of two-phase stirred bioreaction systems by segregated solution of the Euler-Euler model. *Comput. Chem. Eng.* 48, 113–120.
- García-Ochoa, Felix, Gomez, Emilio, 2009. Bioreactor scale-up and oxygen transfer rate in microbial processes: an overview. *Biotechnol. Adv.* 27 (2), 153–176.
- GE Healthcare. Engineering characterization of the single-use Xcellerex XDR-2000 stirred-tank bioreactor system. Application note 29243481AA.
- GE Healthcare Engineering characterization of the single use Xcellerex XDR 500 stirred tank bioreactor system. Application note KA4969250918AN.
- GE Healthcare “Xcellerex™ XDR cell culture bioreactor systems”. Application note KA686051118DF.
- Grenville, R., Giacomelli, J., Padron, G., Brown, D.A.R., 2017. Mixing: Impeller performance in stirred tanks. *Chem. Eng.* 124 (8), 42–51.
- Grenville, R.K., Giacomelli, J.J., VanOmmeren, G.J., Hastings, C.F., Walters, M.J., 2018. Blending in above ground storage tanks with side-entering agitators. *Chem. Eng. Res. Des.* 137, 395–402.
- Grenville, Richard K., 1992. Blending of viscous Newtonian and pseudo-plastic fluids.
- Guo, Z., Shu, C., 2013. *Lattice Boltzmann Method and its Applications in Engineering*, vol. 3. World Scientific.
- Haringa, C., Deshmukh, A.T., Mudde, R.F., Noorman, H.J., 2017. Euler-Lagrange analysis towards representative down-scaling of a 22 m³ aerobic *S. cerevisiae* fermentation. *Chem. Eng. Sci.* 170, 653–669.
- Hinze, J.O., 1955. Fundamentals of the hydrodynamic mechanism of splitting in dispersion processes. *AIChE J.* 1 (3), 289–295.
- Hudcova, V., Machon, V., Nienow, A.W., 1989. Gas-liquid dispersion with dual Rushton impellers. *Biotechnol. Bioeng.* 34 (5), 617–628.
- Kawase, Yoshinori, Halard, Benoit, Moo-Young, Murray, 1992. Liquid-phase mass transfer coefficients in bioreactors. *Biotechnol. Bioeng.* 39 (11), 1133–1140.
- Kawase, Y., Moo-Young, M., 1990. Mathematical models for design of bioreactors: Applications of: Kolmogoroff's theory of isotropic turbulence. *Chem. Eng. J.* 43 (1), B19–B41.
- Krakau, Frederic, Kraume, Matthias, 2019. Three-Dimensional Observation of Single Air Bubble Breakup in a Stirred Tank. *Chem. Eng. Technol.* 42 (7), 1357–1370.
- Krüger, Timm, Halim Kusumaatmaja, Alexandr Kuzmin, Orest Shardt, Goncalo Silva, Erlend Magnus Viggen, 2017. The lattice Boltzmann method. *Springer International Publishing* 10, no. 978-3, pp. 4–15.
- Küng, V.E., Osmanlic, F., Markl, Matthias, Körner, Carolin, 2020. Comparison of passive scalar transport models coupled with the Lattice Boltzmann method. *Comput. Math. Appl.* 79 (1), 55–65.
- Lewis, A.M. et al., 2016. Understanding and controlling sialylation in a CHO Fc-fusion process. *PLoS One* 11 (6), e0157111.
- Lide, David R. (Ed.), 2004. *CRC Handbook of Chemistry and Physics*, vol. 85. CRC Press.
- Mandenius, C.-F., 2016. *Bioreactors: Design, Operation and Novel Applications*. John Wiley & Sons.
- Marques, Marco P.C., Cabral, Joaquim M.S., Fernandes, Pedro, 2010. Bioprocess scale-up: quest for the parameters to be used as criterion to move from microreactors to lab-scale. *J. Chem. Technol. Biotechnol.* 85 (9), 1184–1198.
- Mohamad, A.A., 2011. *Lattice Boltzmann Method*, vol. 70. Springer, London.
- Nasato, Daniel Schiochet et al., 2015. Coarse graining for large-scale DEM simulations of particle flow—an investigation on contact and cohesion models. *Procedia Eng.* 102, 1484–1490.
- Paul, Edward L., Atiemo-Obeng, Victor A., Kresta, Suzanne M. (Eds.), 2004. *Handbook of Industrial Mixing: Science and Practice*. John Wiley & Sons.
- Perry, Robert H., Green, Don W., 2008. *Perry's Chemical Engineers' Handbook*. McGraw-Hill, New York.
- Pope, Stephen B., 2020. *Turbulent flows*. 2020.
- Rathore, A.S., Sharma, C., Persad, A., 2012. Use of computational fluid dynamics as a tool for establishing process design space for mixing in a bioreactor. *Biotechnol. Prog.* 28 (2), 382–391.
- Rettinger, Christoph, Rude, Ulrich, 2018. A coupled lattice Boltzmann method and discrete element method for discrete particle simulations of particulate flows. *Comput. Fluids* 172, 706–719.
- Sajjadi, Baharak, Raman, Abdul Aziz Abdul, Ibrahim, Shaliza, Shah, Raja Shazrin Shah Raja Ehsan, 2012. Review on gas-liquid mixing analysis in multiscale stirred vessel using CFD. *Rev. Chem. Eng.* 28 (2–3), 171–189.
- Sander, R., 2015. Compilation of Henry's law constants (version 4.0) for water as solvent. *Atmos. Chem. Phys.* 15, 4399–4981.
- Sharma, C., Malhotra, D., Rathore, A.S., 2011. Review of computational fluid dynamics applications in biotechnology processes. *Biotechnol. Prog.* 27 (6), 1497–1510.
- Shu, S., Zhang, J., Yang, N., 2020. GPU-accelerated transient lattice Boltzmann simulation of bubble column reactors. *Chem. Eng. Sci.* 214, 115436.
- Shukla, Abhinav A., Gottschalk, Uwe, 2013. Single-use disposable technologies for biopharmaceutical manufacturing. *Trends Biotechnol.* 31 (3), 147–154.
- Succi, Sauro, 2001. *The lattice Boltzmann equation: for fluid dynamics and beyond*. Oxford University Press.
- Sungkorn, R., Derksen, J.J., Khinast, J.G., 2012. Euler-Lagrange modeling of a gas-liquid stirred reactor with consideration of bubble breakage and coalescence. *AIChE J.* 58 (5), 1356–1370.
- Syeda, Sultana R., Ansary, Ashfaq M., 2014. Formation and development of submerged air jets. *J. Mech. Eng.* 44 (2), 137–141.
- To an order-of-magnitude, the time between bubble collisions, can be estimated from kinetic theory as where is the mean bubble diameter, is the gas hold-up, and are the impeller speed and diameter.
- Tuohey, Colin, Dellogio Ted, Erdenberger Thomas, Clapp Kenneth P., Damren Richard, Galliher Parrish M., Guertin Patrick, Lee Jiyoung, Fisher Michael, 2018. Linearly scalable single use bioreactor system. U.S. Patent 9,969,965, issued May 15, 2018.
- Van Leer, Bram, 1979. Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method. *J. Comput. Phys.* 32 (1), 101–136.
- Van't Riet, Klaas, Tramper, Johannes, 1991. *Basic Bioreactor DESIGN*. CRC Press.
- Villiger, Thomas K., Neunstoecklin, Benjamin, Karst, Daniel J., Lucas, Eric, Stettler, Matthieu, Broly, Hervé, Morbidelli, Massimo, Soos, Miroslav, 2018.

- Experimental and CFD physical characterization of animal cell bioreactors: From micro-to production scale. *Biochem. Eng. J.* 131, 84–94.
- Witz, Christian, Treffer, Daniel, Hardiman, Timo, Khinast, Johannes, 2016. Local gas holdup simulation and validation of industrial-scale aerated bioreactors. *Chem. Eng. Sci.* 152, 636–648.
- Xing, Chutian, Wang, Tiefeng, Guo, Kunyu, Wang, Jinfu, 2015. A unified theoretical model for breakup of bubbles and droplets in turbulent flows. *AIChE J.* 61 (4), 1391–1403.
- Yang, Shang-Tian (Ed.), 2011. *Bioprocessing for Value-added Products from Renewable Resources: New Technologies and Applications*. Elsevier.
- Yu, Huidan, Girimaji, Sharath S., Luo, Li-Shi, 2005. DNS and LES of decaying isotropic turbulence with and without frame rotation using lattice Boltzmann method. *J. Comput. Phys.* 209 (2), 599–616.
- Zakrzewski, Roman, 2020. *Simulating the effect of large scale bioreactor heterogeneities on cell culture performance: A combined experimental and modelling approach*. PhD diss., University College London.
- Zanghi, Adiba, Debora Lin, Tina Balsara, Lucian Young, Barry Wolf, Ping Huang, 2017. Increased efficiency and product quality with the UniVessel® Single Use bioreactor for CHO fed-batch cultures.
- Zhu, Luoding, He, Guowei, Wang, Shizhao, Miller, Laura, Zhang, Xing, You, Qian, Fang, Shiao-fen, 2011. An immersed boundary method based on the lattice Boltzmann approach in three dimensions, with application. *Comput. Math. Appl.* 61 (12), 3506–3518.